

Student Ability Modeling

Prompt

You are an expert in Educational Data Mining and Knowledge Tracing.

Task: Conduct a comprehensive assessment of the student's knowledge proficiency by integrating their Historical Interaction Logs and Emotion Probability Distributions.

Based on the input data, perform the following two tasks:

- 1. Performance Prediction: Predict the probability of correctness for the "Next Question" related to the target knowledge concept. (Range: 0.00 – 1.00 rounded to 2 decimal places).
- 2. Proficiency Scoring: Estimate a comprehensive knowledge proficiency score. (Range:* 0 – 100 Integer). Rationale, Provide a concise 1-2 sentence justification.

Output Format: Strictly follow this format.

Probability: <Value 0.00-1.00>

Proficiency Score: <Integer 0-100>

Rationale: <Brief explanation>

Student ID: 33

Historical Interactions (Chronological Order, Last 20):

Seq	Skill	Correctness	Time Taken	Hint Usage
1	noskill	0	12.0 s	1
2	noskill	1	12.0 s	0
3	noskill	1	13.0 s	0
4	area	1	16.0 s	0
5	noskill	0	16.0 s	0
6	noskill	1	20.0 s	0
7	noskill	0	27.0 s	0
8	noskill	1	31.0 s	0
9	pythagorean-theorem	1	34.0 s	0
10	noskill	1	36.0 s	0
11	noskill	1	55.0 s	0
12	venn-diagram	1	56.0 s	0
13	noskill	1	74.0 s	0
14	area	1	78.0 s	0
15	equation-concept	1	147.0 s	0
16	noskill	1	177.0 s	0
17	area	0	179.0 s	0
18	noskill	0	183.0 s	0
19	noskill	0	201.0 s	0
20	noskill	1	430.0 s	0

Affective State Profile (Fixed Probabilities):

- Carelessness: 0.20
 - Boredom: 0.26
 - Engagement/Concentration: 0.65
 - Confidence: 0.06
 - Frustration: 0.06
 - Off-task: 0.30
 - Gaming the System: 0.01
- Target Prediction Task:
- Next Knowledge Concept: noskill



Probability: 0.75

Proficiency Score: 66

Rationale: The student demonstrates consistent performance on 'noskill' items, maintaining an accuracy of approximately 60%. Despite the extended response times, the overall correctness remains satisfactory.